

PRACTICE 2: PSD of Random Signals

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GitHub: https://github.com/javierba98/Com2_A1_G9.git

Abstract

This report is about the practical analysis of signals in time and Power Spectral Density (PSD). It also covers the use of simulation software (GNURadio) for signal processing. Different types of signals were studied, including bipolar random signals, binary signals, audio signals, and images. These signals were analyzed for their spectral content, bandwidth, and other specific parameters.

I. INTRODUCTION

In the field of communications, signal analysis is essential. This report deepens the understanding of different types of signals, including bipolar random signals, binary signals, audio, and images. Using the GNURadio simulation software, the behavior of these signals is observed in both the time and frequency domains. Additionally, important topics in communications, such as interpolation and signal modulation, are covered. This provides a better understanding of how to analyze and manipulate signals in various applications.

II. THEORETICAL FRAMEWORK

The knowledge of PSD of random signals is essential for various applications in communications. Its study allows for noise analysis, signal processing, and the analysis of natural phenomena, which is key in the characterization of signals in telecommunications. To better understand its operation, the theoretical definitions used for this topic will be covered.

- **PSD:** It is a mathematical function that shows how the power of a signal is distributed across different frequencies. This function helps analyze the power present in each frequency component, providing a detailed representation of how the signal's energy is spread in the frequency domain [1].

- **Bipolar Random Binary Signal:** It is a way of representing digital data where each bit can take one of two possible values, typically '0' or '1'. In bipolar encoding, these bits are represented using three voltage levels: positive, negative, and zero. Specifically, '1's alternate between positive and negative levels, while '0's are represented by a zero voltage level [2].
- **Bit Rate:** It refers to the number of bits transmitted per second [3].
- **Sampling Frequency:** It refers to the number of samples taken per unit of time from a continuous signal to produce a discrete signal [4].
- **Bandwidth:** Bandwidth is the capacity of a communication link to transmit data from one point to another within a specific time period, usually measured in bits per second (bps) [5].
- **White Noise:** It is a random signal, known as a stochastic process, characterized by the fact that the signal values at different times have no statistical correlation. As a result of this property, the PSD remains constant [6].

III. METHODOLOGY

To carry out the laboratory, research on the topic was conducted using the provided guide. Tools such as GitHub, Linux, and Moodle were used for development and process management. Each stage was documented in detail to ensure the reliability of the results, following these steps:

- Step 1 - Environment Preparation:** The proposed flowchart for the practice was verified by analyzing a bipolar random binary signal with a rectangular shape. Additionally, the signal was characterized by determining its main parameters, which are presented in the following table.

h	sps	rate bit	samp rate	bandwidth
1, 1, 1, 1	4	32k	128k	64kHz
1, 1, 1, 1, 1, 1, 1, 1	8	32k	256k	64kHz
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1	16	32k	512k	64kHz
1	1	32k	32k	64kHz

Table 1. Signal Parameters According to h

- b. Step 2 - Analysis of White Noise in the Time Domain and PSD:** The necessary blocks were configured in GNURadio to visualize white noise in both the time domain and its Power Spectral Density (PSD). The required parameters were adjusted to ensure an appropriate representation, and the results are presented in the following graphs.

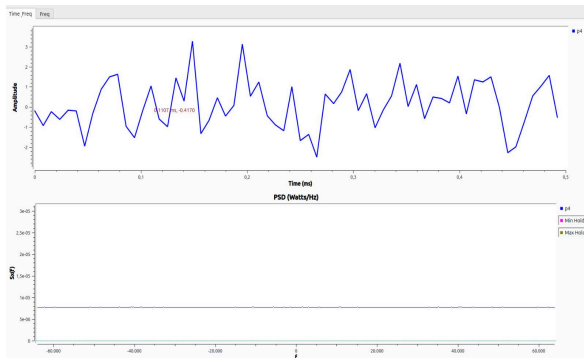


Fig. 1. White Noise and Its PSD

- c. Step 3 - Analysis of the Signal from a Real-World Source:** The behavior of the signal in the time and frequency domains was analyzed when the bits were obtained from real sources, such as an image and an audio file. To achieve this, the source block was configured in GNURadio, allowing the observation of how the signal varies depending on different inputs and their corresponding spectral representation.

To carry out this analysis, tests were first conducted using an image source. Various images with different shades and textures were evaluated, allowing observation of how these characteristics affect the processed signal.

Test 1:



Fig. 2. Frog Image

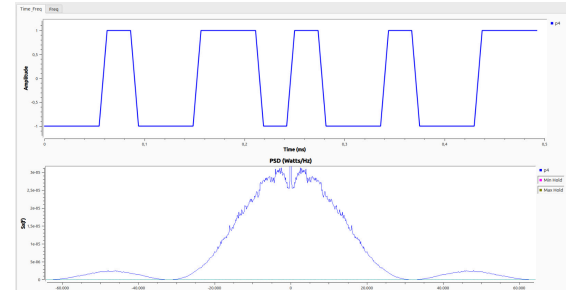


Fig. 3. Frog Signal in the Time Domain

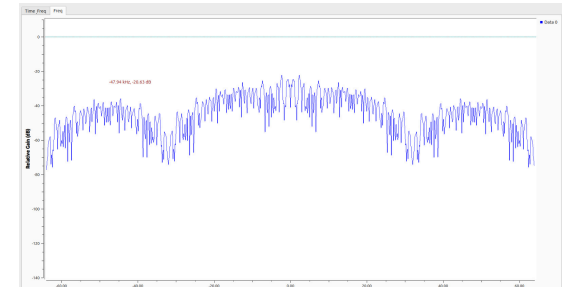


Fig. 4. Frog Signal in the Frequency Domain

Test 2:

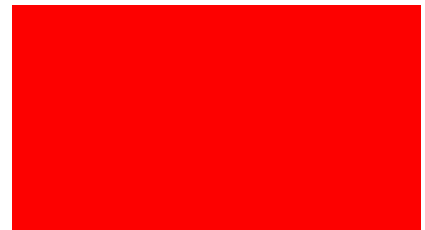


Fig. 5. Red Background

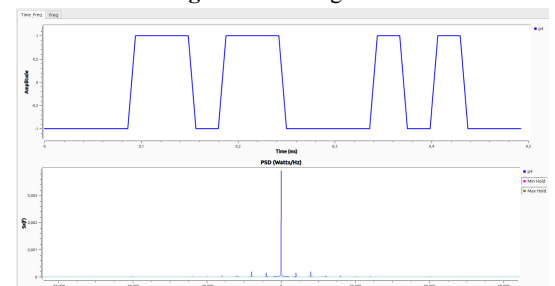


Fig. 6. Red Background Signal in the Time Domain

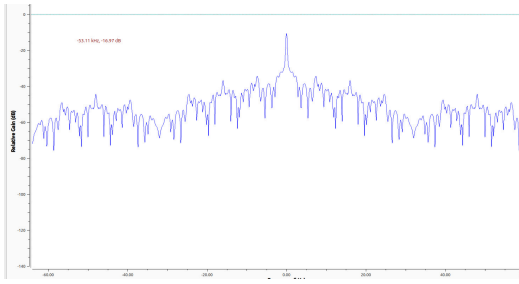


Fig. 7. Red Background Signal in the Frequency Domain

Test 3:



Fig. 8. Rainbow Image

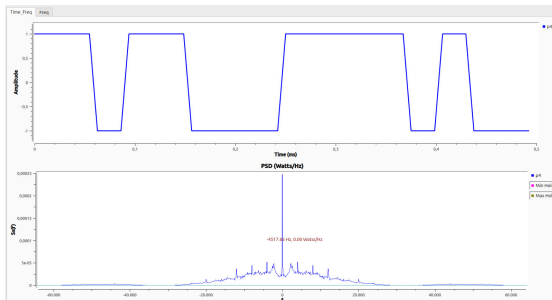


Fig. 9. Rainbow Signal in the Time Domain

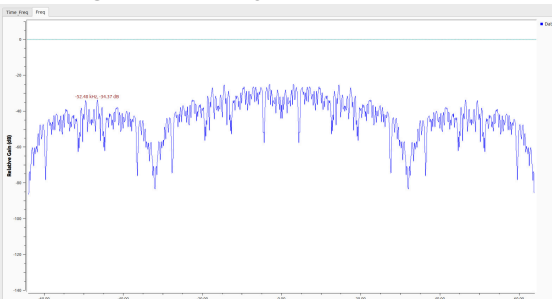


Fig. 10. Rainbow Signal in the Frequency Domain

In the second part of the process, the procedure was repeated using an audio signal source, allowing the analysis of its behavior in both the time and frequency domains.

Test 1 - Test Audio:

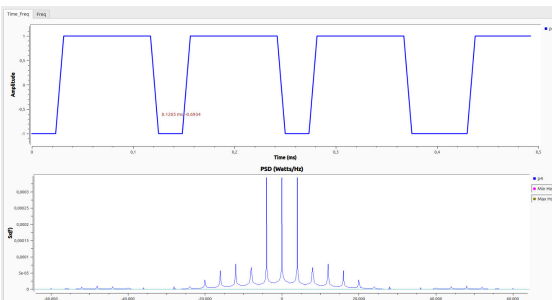


Fig. 11. Test Audio Signal in the Time Domain

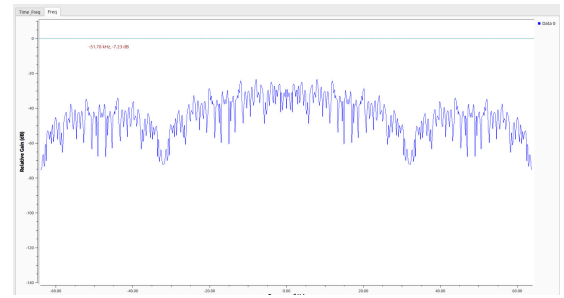


Fig. 12. Test Audio Signal in the Frequency Domain

Test 2 - Relaxing Music for Sleep:

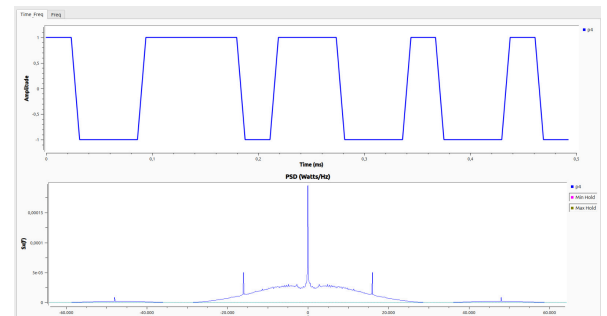


Fig. 13. Relaxing Music Signal in the Time Domain

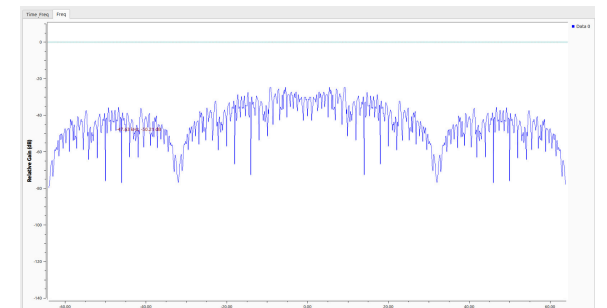


Fig. 14. Relaxing Music Signal in the Frequency Domain

Test 3 - Electronic Music:

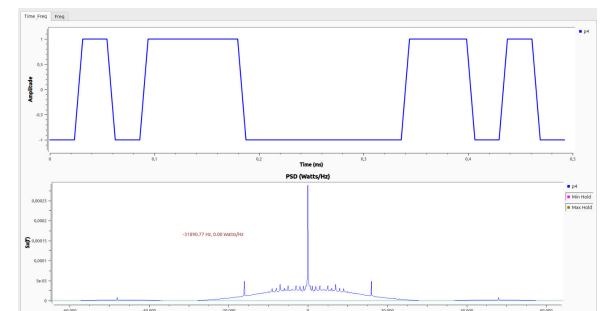


Fig. 15. Electronic Music Signal in the Time Domain

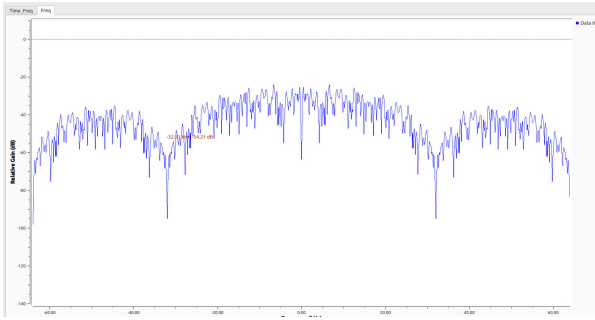


Fig. 16. Electronic Music Signal in the Frequency Domain

IV. ANALYSIS OF RESULTS

Based on the data obtained in the experiment, a detailed analysis was conducted to interpret the signal's behavior in different scenarios. This section presents the main findings and their significance within the context of the experiment.

a. Part 1 - Environment Preparation:

To analyze this section, tests were performed, and the collected data were summarized in Table 1. This table shows various signal characteristics for different values of Sps (samples per symbol). It was observed that the variation of Sps does not affect the bit rate or the signal bandwidth but does influence the sampling frequency.

The bit rate and bandwidth remain unchanged with variations in Sps because these parameters depend on the number of symbols transmitted per second and the type of modulation used, rather than the number of samples taken per symbol.

Specifically, an increase in the sampling frequency is evident as Sps increases, following a pattern where the selected values are multiples of two, except when $Sps = 1$, in which case it must be multiplied by 4. Additionally, it is confirmed that as Sps grows, the sampling frequency also increases since more samples of the signal are being taken.

b. Part 2 - Analysis of White Noise in the Time Domain and PSD

Image Tests: First, the signal behavior in the time domain was analyzed, where it was observed that the signals are bipolar and random. Regarding its PSD, for Image 1 (frog photo), the power spectral density is distributed according to

the variations in intensity and texture of the image.

The PSD bandwidth is relatively wide due to the diversity of tones and transitions present in the image. Although green tones dominate, they vary in intensity, with darker and lighter regions. The PSD shows that the most relevant spectral components are associated with frequent variations in the image, while the lower-energy frequencies correspond to areas with less variability in tones and textures.

For Image 2 (red background), the signal in the time domain does not exhibit abrupt transitions, as the image consists of a single uniform color. As a result, the dominant frequency corresponds to the red color, generating a strong DC component, as shown in Figure 6. Additionally, the spectral characteristics of the signal are concentrated in low frequencies, and the bandwidth is not wide since the absence of color variations reduces the presence of high-frequency components in the PSD, making it more uniform.

Finally, in the rainbow image, the PSD shows a more extended frequency spectrum compared to the red background case. There are multiple components at different frequencies, confirming the presence of rapid transitions in the image. Although the rainbow image has more color changes than the frog image, its PSD does not have the same shape, as the color transitions are relatively evenly distributed throughout the image.

In contrast, in the frog image, there are areas with predominant colors and well-defined edges, which generate more specific peaks in the PSD, associated with these transitions. In other words, the frog image has a more complex structure with more localized changes, while the rainbow background presents more global and uniform color transitions throughout the image.

Secondly, the differences between the three image tests in the frequency domain will be analyzed. These differences are similar to those observed in the time domain since, in the image with a red background, the frequency components tend to remain stable across all frequencies. This is because the image does not exhibit significant color changes, resulting in a uniform distribution in the spectrum.

On the other hand, in the frog and rainbow images, "hill-like" patterns can be observed, indicating the

presence of periodicity in certain color transitions. In the case of the rainbow, the colors change relatively uniformly throughout the image rather than in a completely random manner. This results in more global and distributed transitions, without a single dominant frequency. Instead, the image presents multiple frequency ranges with higher energy, suggesting color variations at different transition scales.

Audio Tests: In the case of the time-domain graphs, it can be observed that in the test audio, the PSD peaks in Figure 11 are very pronounced. This is due to the presence of abrupt changes between the audio tones.

On the other hand, in the case of the sleep music audio, the PSD shows a much smoother variation, without abrupt changes in its shape. This is because this type of music is designed to be relaxing, so its tones are more stable, as evidenced in the PSD of Figure 13.

Finally, in the test with electronic music, the PSD shows small changes, indicating variations in sound intensity. However, although this type of music is characterized by strong sounds, they tend to remain constant or have a relatively stable tone. Regarding the bandwidth, all tests show similar values.

Regarding the frequency domain, the graphs tend to look quite similar since these types of music, although different, contain tones at various frequencies. As a result, similar frequencies group together, creating the "hill-like" effect. Clearly, if the transitions in the PSD are more abrupt, this indicates that the transition in time is also abrupt. An example of this can be seen in the peaks and changes between electronic music and lyrical music.

Control Question

a. What role does the following block combination play?

This block combination in GNU Radio applies a transformation to the input signal. First, the Constant Source block generates a constant value of -0.5, which is added to the signal in the Add block, shifting it downward. Then, the Multiply Const block multiplies the result by 2, scaling the signal.

b. What role does the "Interpolation FIR Filter" block play, and how does it work?

The Interpolation FIR Filter block in GNU Radio increases the sampling rate of a signal through interpolation. Its main function is to insert additional samples between the original ones.

c. Why is the PSD of binary signals derived from an audio signal different from that of a photo, even though both are bipolar rectangular binary signals?

The difference in the PSD of binary signals from audio and images is due to the nature of the information they represent. Audio is a continuous signal with smooth variations, leading to a spectral distribution concentrated in certain bands. In contrast, an image converted to binary has abrupt transitions depending on its visual content, introducing more high-frequency components. Additionally, the encoding method influences the transition statistics, with smoother transitions in audio and denser transitions in images with many details.

d. What role does the "Throttle" block play?

The Throttle block in GNU Radio is responsible for limiting the signal processing speed when there is no external hardware controlling the data flow. Without this block, the flow execution could consume all processor resources, as GNU Radio processes data as fast as possible. Throttle imposes a specific sampling rate, ensuring efficient CPU usage and enabling a more realistic simulation of real-time signals.

e. What would happen to the PSD if the signal is not converted to a bipolar signal, and the binary signal in p4 only has values of 0 or 1 instead of -1 or 1?

If the signal is not converted to a bipolar format, the Power Spectral Density (PSD) would change significantly. A unipolar signal has a DC component because its values are only positive or only negative, leading to a concentration of energy at zero frequency. In contrast, converting it to bipolar removes this DC component, redistributing the energy across the spectrum, resulting in a more balanced PSD centered on the relevant frequencies.

f. White noise is assumed to have infinite bandwidth. Does this match what is observed in GNU Radio? Why?

It depends on the perspective of analysis. It could be considered to have infinite bandwidth, as it intersects the horizontal axis, or no bandwidth at all, since it appears as a constant. This matches what was observed in the experiment.

g. A random rectangular binary signal is theoretically assumed to have infinite bandwidth. Does this match what is observed in GNU Radio? Why?

In theory, a random rectangular binary signal has infinite bandwidth due to its discontinuous transitions, which introduce high-frequency components. However, in GNU Radio, the observed bandwidth is limited due to practical

constraints such as sampling, filtering, and system resolution. These factors attenuate higher frequencies, causing the PSD to display a finite spectrum instead of a truly infinite one.

h. What formula could help calculate the number of lobes in the PSD of a random rectangular binary signal when the sampling frequency and Sps are known?

To calculate the number of lobes in the Power Spectral Density (PSD) of a random rectangular binary signal, we can use the relationship between the sampling frequency (f_s), the samples per symbol (Sps), and the signal bandwidth.

A rectangular binary signal has a spectrum following a $\text{sinc}^2(f)$ function, where the main lobe width is determined by the symbol rate (f_b), given by:

$$f_b = f_s / \text{Sps}$$

Since the lobes in the PSD appear at multiples of f_b , the total number of visible lobes (N) in the spectrum up to the Nyquist frequency can be approximated as:

$$N = f_s / 2 * f_b + 1 = \text{Sps} / 2 + 1$$

The +1 term is due to the central lobe, which counts as two due to its double width.

i. How is the total frequency range of the spectrum calculated when Sps and Rb are known?

The total frequency range occupied by the spectrum is defined as the bandwidth displayed in the GNU Radio simulator. This parameter is determined by the sampling frequency. Therefore, knowing R_b and Sps, the range is:

$$f_s = R_b * \text{Sps}.$$

j. How is the spectral resolution of the spectrum analyzer calculated when N and the sampling frequency are known?

The spectral resolution (Δf) of a spectrum analyzer based on the Discrete Fourier Transform (DFT) is determined by the number of FFT points (N) and the sampling frequency (f_s). It is calculated using the following equation:

$$\Delta f = f_s / N$$

k. What would happen if the parameter K in the "Unpack K Bits" block is set to 16?

The variability of the transmitted symbols increases, and its PSD contains more frequency components with a wider power distribution in the frequency domain.

l. How would you calculate the sampling frequency at the input of the "Unpack K Bits" block if the number of PSD lobes and the signal bandwidth are known?

Bit packing is directly related to the bit rate (R_b). Given the bandwidth and the number of lobes (Sps), along with the equations shown below, it can be determined that the sampling frequency at the output of the "Unpack K Bits" block is the same as the bit transmission rate (R_b).

$$BW = \text{Samprate}(p4) = R_b * \text{Sps}$$

$$\text{Samprate}(p3) = \text{Samprate}(p4) / \text{Sps}$$

m. How would you calculate the sampling frequency at the output of the "Unpack K Bits" block if the input sampling frequency is known?

The output sampling frequency is the input sampling frequency multiplied by the number of bits per packet (K).

n. How would you calculate the sampling frequency at the output of the "Char to Float" block if the input sampling frequency is known?

The output sampling frequency of the "Char to Float" block is the same as in the previous point, i.e., $f_s * K$, since this block only converts character-type data to floating-point type, and the input sampling frequency remains unchanged at the output.

o. For what value of Sps does the PSD of a bipolar random binary signal resemble the PSD of white noise?

The PSD of a bipolar random binary signal resembles that of white noise when the number of samples per symbol (Sps) is much greater than 1, typically $\text{Sps} \geq 8$. This is because a high Sps spreads the signal's energy over a wider band, making its spectrum more uniform and less affected by the typical sinc^2 function decay, thus making it resemble white noise over a broad frequency range.

p. What minimal changes would you make to the flowchart, mainly manipulating h , if you want the bits in the random binary signal to take the shape of sawtooth waves?

To obtain a sawtooth wave signal, the array in the variable h was replaced with a list created using the

np.arrange function. Additionally, the value of the constant source block was set to 0, and the input was left with the random block.

q. What minimal changes would you make to the flowchart, mainly manipulating h, if you want the random binary signal to have Unipolar RZ line coding?

To achieve Unipolar RZ line coding, the size of the vector h was increased to take a completely square shape. Additionally, the constant source block was set to a value of 0 so that the signal appeared as expected.

r. What minimal changes would you make to the flowchart, mainly manipulating h, if you want the random binary signal to have Manchester NRZ line coding?

For Manchester coding, the vector h was manipulated with the corresponding encoding so that the random binary signal passed through the filter and behaved as follows: each bit 1 corresponds to a transition from low to high, and each bit 0 corresponds to a transition from high to low.

s. What minimal changes would you make to the flowchart, utilizing h and the FIR Interpolating Filter, to shape the binary signal into an OOK signal?

To achieve an output signal that meets the expected characteristics of OOK modulation, the array in h was replaced with a sinusoidal signal concatenated with a constant value of 0. The np.concatenate and np.arrange functions were used for this purpose.

t. What minimal changes would you make to the flowchart, utilizing h and the FIR Interpolating Filter, to shape the binary signal into a BPSK signal?

To obtain BPSK modulation at the output, the array in the variable h was replaced with two concatenated sinusoidal functions—one with negative amplitude and the other with positive amplitude—to achieve the phase shift characteristic of this modulation. Additionally, to efficiently analyze the result, the input was changed to an infinitely repeating vector alternating between 0 and 1.

u. What minimal changes would you make to the flowchart, utilizing h and the FIR Interpolating Filter, to shape the binary signal into an ASK signal?

The output signal took the form of ASK modulation by replacing h with a sinusoidal function. Additionally, to make the results easier to analyze, the random input was replaced with a predefined vector.

v. What minimal changes would you make to the

flowchart, utilizing h and the FIR Interpolating Filter, to shape the binary signal into a heartbeat-like signal?

The output signal took the shape of a heartbeat by replacing h with a sinc(x) function. To make it resemble the expected shape, adjusting the step values was crucial. The random input was replaced with a vector alternating between 0 and 1 to obtain a more natural response.

w. What minimal changes would you make to the flowchart, utilizing h and the FIR Interpolating Filter, to shape the binary signal as shown in Figure 7?

To obtain the rippled pulses, the NumPy function np.concatenate was used, adding a negatively scaled decaying exponential, followed by an array of ones, and another positively scaled decaying exponential. The random input was replaced with a predefined vector, resulting in a graph very similar to the desired one.

x. Explain, using PSD graphs, the difference between the PSD of a bipolar binary signal and a unipolar binary signal.

The main difference between the PSD of a bipolar and a unipolar binary signal is that the unipolar signal has a DC component due to its nonzero mean, whereas the bipolar signal does not, as its mean is zero. Additionally, the PSD of the bipolar signal distributes its energy more effectively across the spectrum, reducing concentration at low frequencies and improving spectral efficiency. In contrast, the unipolar signal concentrates more energy at DC and low frequencies, which can cause issues in certain transmission system [9]s.

V. CONCLUSIONS

- **The power spectral density is useful for** analyzing how the energy of a signal is distributed in the frequency domain. It helps identify dominant spectral components, detect noise, and optimize the design of communication and signal processing systems.
- **Images with more color transitions and loud audio in the frequency and time domain** tend to have more noticeable variations in their graphs. This means they have higher frequency content and greater spectral complexity, which helps in compression and digital processing applications.
- **The interpolation block takes the input signal with a certain sampling rate and interpolates it** to generate extra samples. This helps to preserve information during the process, reduce

aliasing, and improve the quality of signal reconstruction.

- **The GNURadio program helped with these practices because it has blocks** that allow real-time signal simulation and processing, making it easier to implement and analyze communication systems without the need for specialized hardware

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